



The chip from University of Texas at San Antonio (UTSA) makes batteries smaller and low-power electronics work more efficiently (Image courtesy: www.utsa.edu)

the size of a pin's head, but has great potential in applications ranging from everyday electronics to medical devices.

The chip can keep a phone working at top functionality with much less power. Moreover, it facilitates the use of smaller batteries, since the object itself is so small. The chip also tackles another common annoyance for electronics users: how hot devices get when they're being used for several minutes.

"The heat is a result of a lot of power being used," said Ruyan Guo, team leader and professor of Electrical and Computer Engineering. "With our device there is less power consumption, which means the heat will be much less of an issue."

Heat-resistant capacitors

Fraunhofer researchers have developed a capacitor that can withstand temperatures of up to 300 degrees Celsius. By comparison, conventional electronics can cope with temperatures only up to 125 degrees.

Researchers did so by using an innovative mix of materials—and a special 3D trick. When manufacturing the conducting metal layers of the capacitor, the team etched tiny holes in the surface to increase its area. This 3D trick increases its capacity and simultaneously makes it possible to use a thicker dielectric. A thicker layer withstands higher temperatures better and can decrease uncontrolled leakage current in the capacitor.

The experts adopted a new mix of materials in the production of the insulating dielectric too by using tantalum



Heat-resistant capacitor developed by Fraunhofer researchers can withstand temperatures of up to 300 degrees Celsius (Image courtesy: www.fraunhofer.de)

pentoxide—a compound of tantalum (a metal) and oxygen and aluminium oxide. Tantalum pentoxide is better at storing charge carriers than the silicon oxide usually used, so it increases the capacitor's capacitance. In addition, the Fraunhofer researchers used a highly electrically conductive silicon as well as ruthenium, which is especially robust and heat-resistant.

IoT systems usually link networks of sensors via radio, but radios demand battery power thus limiting usability. Disney Research has determined that one solution may be to get rid of the radios all together and communicate via the ambient radio waves from TV, radio and cellphones.

Search engine for 3D printable models

What Google does for searching information on the Internet, new search engine IFind3D will do for printable 3D models. Launched by 3D Ninja, the largest reseller of 3D printers in The Netherlands, www.ifind3d.com contains over 700,000 designs. It combines online libraries for 3D printable models in one fast and responsive engine, making it easy for end users to find 3D printable models.

At the moment, almost 70 per cent of all online libraries and repositories are connected. It is expected that before the year-end around 90 per cent of all worldwide 3D printable designs will be indexed in the search engine.

IFind3D uses state-of-the-art algorithms to make sure the best results are served at all time using almost 100 different variables by integrating an AI service using IBM's supercomputer 'Watson.'

Pivotal technologies for IoT applications

ON Semiconductor has developed a variety of flexible, energy-efficient, high-performance solutions for emerging Internet of Things (IoT) applications.

The new AR0237 RGB-IR and AR0238 RGB-IR CMOS image sensors can capture daytime colour and nighttime near-infrared (nIR) image data on the same sensor without the cost and complexities (like refocusing and maintenance) of a mechanical IR-cut filter mounted onto the imaging assembly. Their unique colour filter array (CFA) arrangement employs a 4x4 kernel, which replaces some red and blue pixels with nIR-sensitive pixels and rearranges the spatial density of the remaining pixels. Thus, the devices can deal with the most challenging of lighting conditions. These devices are highly suited to home security and automated monitoring applications where ambient light levels vary greatly over time.

From a connectivity perspective, the company has developed the RSL10 multi-protocol Bluetooth 5 certified radio system-on-chip. Claimed to offer the industry's lowest power consumption in peak receiving and deep sleep mode, the RSL10 is optimised for use in IoT edge-node devices, as well as for health and wellness applications.